

## Assignment: module- 5

### Network Fundamentals and Building Networks Section

#### Section 1: Multiple Choice

1. What is the primary function of a router in a computer network?
  - a) Assigning IP addresses to devices
  - b) Providing wireless connectivity to devices
  - c) **Forwarding data packets between networks**
  - d) Managing user authentication and access control
  
2. What is the purpose of DHCP (Dynamic Host Configuration Protocol) in a computer network?
  - a) Assigning static IP addresses to devices
  - b) Resolving domain names to IP addresses
  - c) Managing network traffic and congestion
  - d) **Dynamically assigning IP addresses to devices**
  
3. Which network device operates at Layer (Data Link Layer) of the OSI model and forwards data packets based on MAC addresses?
  - a) Router
  - b) **Switch**
  - c) Hub
  - d) Repeater
  
4. Which network topology connects all devices in a linear fashion, with each device connected to a central cable or backbone?
  - a) Star
  - b) **Bus**
  - c) Ring
  - d) Mesh Section

## Section 2: True or False

5. A VLAN (Virtual Local Area Network) allows network administrators to logically segment a single physical network into multiple virtual networks, each with its own broadcast domain.
  - **TRUE** (A VLAN divides one physical network for better security and performance.)
6. TCP (Transmission Control Protocol) is a connectionless protocol that provides reliable, ordered, and error-checked delivery of data packets over a network.
  - **FALSE** (Tcp is connection based and ensures data is sent Correctly and in order.)
7. firewall is a hardware or software-based security system that monitors and controls incoming and outgoing network traffic based on predetermined security rules.
  - **TRUE** (A firewall protects a network by controlling and checking data traffic.)
8. Describe the steps involved in setting up a wireless network for a small office or home office (SOHO) environment.

Setting up a wireless network at home or in a small office is simple if you follow the steps carefully. It allows multiple devices like laptop, phones, and printers to connect to the internet without cables.

1. Select good wireless router that supports modern Wi-Fi standard.
2. Connects the router and wait for it to boot up.
3. Turn on the router and wait for it boot up.
4. Open the router's setup page by typing its IP address (for example, 192.168.1.1) in a browser.
5. Login using name (SSID) and a strong password.

6. Enable DHCP so that router gives IP addresses automatically.
7. Place the router in a central location for better signal coverage.
8. Connect devices using the Wi-Fi name and password.
9. Test the internet connection on multiple devices.

## **Section 4: Practical**

9. Demonstrate how to configure a router for Internet access using DHCP (Dynamic Host Configuration Protocol).

To configure a router for internet access using DHCP, follow these steps:

1. **Connect your router and computer:**

Plug your router into a power source and connect it to your computer using a LAN cable or Wi-Fi.

2. **Access the router's setup page:**

Open a web browser on your computer and type the router's IP address in the address bar (for example, 192.168.1.1 or 192.168.0.1), then press enter.

3. **Login to the router:**

you will be asked to enter a username and password. These are usually printed on the back of the manual (common ones are admin/ admin or admin/password).

4. **Go to internet or WAN setting:**

In the router's menu, find a section called “**internet setting**”, “**network setting**”, or “**WAN Setup**”.

5. **Select the connection type:**

Form the list of connection option, choose “DHCP” OR “Automatic IP.” This tells the router to automatically get its IP address and other network details from the internet services provider (ISP).

6. **Save the setting:**

After selection DHCP, click on SAVE OR APPLY to confirm the changes.

**7. Restart the router:**

Some routers restart automatically after saving; if not, manually turn it off and back on to apply the new setting.

**8. Check internet connection:**

Once the router restarts, open the **status** or **network Information** page. You should see an IP address assigned by the ISP, which means the internet is working.

**9. Test the connection:**

Connect your mobiles or laptop to the Wi-Fi and try opening any website to confirm that the internet connection is active.

## **Section 5: Essay**

**10. Discuss the importance of network documentation in the context of building and managing networks.**

Network documentation means keeping a complete and organized record of all the details related to a computer network. This includes diagrams, IP address lists, device names, router and switch configuration, cable connection, username, cable connection, username, password, and other important settings. In short, it is a written or digital record that explains how the network is built and how it works.

When an organization or institution builds a computer network, proper documentation is one of the most important steps. It ensures that everyone who manages or uses the network can understand its structure clearly. Without documentation, it can be very difficult to find a problem or make changes in the future.

The main reasons why network documentation is so important:

**1. Helps in Troubleshooting problem:**

If a network problem happens, documentation Helps identify where the issue might be. For example, if a device is not connecting to the internet, the network diagram can show which router or switch it is connected to. This saves time and helps solve the problem faster.

**2. Saves time and effort:**

When everything is recorded properly, there is no need to guess or search for network details. Technicians or network administrators can refer to the documentation to quickly understand the setup and take the right action.

**3. Improves network security:**

Network documentation helps keep tracks of which devices and users have access to the network. It also stores information about firewalls, passwords, and security policies. This makes it easier to spot any security risk and take preventive measures.

**4. Helps during upgrades or expansion:**

when company want to expand or upgrade its network, documentation helps a lot. It shows which parts can be upgraded and where new devices can be added. This reduces confusion and prevents errors during installation.

**5. Useful for new staff or administrators:**

when new employees or IT staff join, they can easily understand the network structure by reading the documentation. They do not need to depend completely on need to depend completely on other people for information.

**6. Ensures consistency and standardization:**

Documentation helps maintain consistency in naming devices, assigning IP addresses, and configuring network setting. This makes the network organized and easier to manage.

**7. Supports compliance and auditing:**

Many organizations must follow government or company security rules. During audits, network documentation provides proof of how the network is managed and protected. This is important for passing compliance checks.

**8. Reduces downtime:**

When a network issue occurs, having proper documentation helps proper documentation helps proper documentation helps restore the system quickly because all configuration and devices details are available. This reduces business downtime and keeps users productive.

Basically, network documentation is like a “map” or “manual” for the entire network. It helps in understanding how everything is connected and how to fix or improve it when needed. Well-maintained documentation saves time, increases efficiency, strengthens security, and ensures that the network can grow smoothly in the future. For these reasons, every organization should treat network documentation as an essential part of building and managing its network.